DNSSEC FLAGS AND FIELDS - QUICK REFERENCE

(bit 0 = most significant bit, as RFCs number them)

1. DNS MESSAGE HEADER FLAGS (RFC 1035, RFC 4035)

Flag	Bit	Set by	Meaning
QR	0	responder	0 = query, 1 = response
Opcode	1-4	client	0 QUERY, 2 STATUS, 4 NOTIFY, 5 UPDATE
AA	5	authoritative server	Authoritative Answer
TC	6	responder	TrunCated: retry over TCP
RD	7	client	Recursion Desired
RA	8	resolver	Recursion Available
Z	9	-	reserved, must be 0
AD	10	validating resolver	Authentic Data: validated as secure
CD	11	client	Checking Disabled: "don't validate"
RCODE	12-15	responder	response code (section 7)

2. EDNS(0) OPT RECORD FLAGS (RFC 6891, RFC 3225, RFC 9824)

Flag	Mask	Meaning
DO	0x8000	DNSSEC OK: include RRSIG/NSEC/NSEC3/DNSKEY records
CO	0x4000	Compact answers OK (RFC 9824): client accepts compact denial (NXNAME) and wants NXDOMAIN restored
Z	rest	reserved, must be 0

Other OPT fields: version (0; else BADVERS), UDP payload size (e.g. 1232), extended RCODE (upper 8 bits of the rcode).

3. DNSKEY FLAGS FIELD (RFC 4034, RFC 5011)

Flag	Bit	Value	Meaning
ZONE	7	256	zone key; must be set for DNSSEC signing
REVOKE	8	128	key revoked (RFC 5011 rollover)
SEP	15	1	Secure Entry Point: marks a KSK (hint only)

Common values: 256 = ZSK, 257 = KSK or CSK (ZONE+SEP), 385 = revoked KSK
Protocol field: always 3.

4. ALGORITHM NUMBERS (DNSKEY, RRSIG, DS)

5	RSASHA1	deprecated
7	RSASHA1-NSEC3-SHA1	deprecated
8	RSASHA256	common (root zone)
10	RSASHA512	
13	ECDSAP256SHA256	common (.com, Cloudflare, this project)
14	ECDSAP384SHA384	
15	ED25519	
16	ED448	

Retired or rare: 1 RSAMD5, 3 DSA, 6 DSA-NSEC3-SHA1, 12 ECC-GOST
DS digest types: 1 SHA-1 (deprecated), 2 SHA-256, 3 GOST R 34.11-94, 4 SHA-384

5. NSEC3 / NSEC3PARAM FIELDS (RFC 5155, RFC 9276)

Hash algorithm	1 = SHA-1 (the only one)
Flags (NSEC3)	bit 7 = Opt-Out (value 1): span may contain unsigned delegations; flags other than 0/1 -> record ignored
Flags (NSEC3PARAM)	must be 0
Iterations	extra hash rounds; RFC 9276 says 0
Salt	"-" = none; RFC 9276 says none

6. DNSSEC RECORD TYPES

DS 43 RRSIG 46 NSEC 47 DNSKEY 48 NSEC3 50 NSEC3PARAM 51
CDS 59 CDNSKEY 60 NXNAME 128 (compact denial, RFC 9824)

RRSIG fields: type covered, algorithm, labels (owner labels not counting a leading "*"; fewer than the owner's labels = wildcard expansion), original TTL, expiration, inception, key tag, signer name, signature.

7. RESPONSE CODES (RCODE)

```
 0  NOERROR    answer, or NODATA (empty answer)
 1  FORMERR    malformed query
 2  SERVFAIL   resolver could not resolve; DNSSEC bogus
 3  NXDOMAIN   name does not exist
 4  NOTIMP     not implemented
 5  REFUSED    policy (not our zone, AXFR)
16  BADVERS    unsupported EDNS version
```

8. EXTENDED DNS ERRORS, DNSSEC-RELATED (RFC 8914, carried in OPT)

```
 1  Unsupported DNSKEY Algorithm     8  Signature Not Yet Valid
 2  Unsupported DS Digest Type       9  DNSKEY Missing
 5  DNSSEC Indeterminate            10  RRSIGs Missing
 6  DNSSEC Bogus                    11  No Zone Key Bit Set
 7  Signature Expired               12  NSEC Missing
                                    27  Unsupported NSEC3 Iterations Value
(Codes written from memory: check RFC 8914 / the IANA registry.)
```

9. VALIDATION OUTCOMES

```
Secure         chain from a trust anchor; AD set
Insecure       provably unsigned; no AD
Bogus          validation failed -> SERVFAIL
Indeterminate  no trust anchor covers the name
```

10. TOOL CHEAT SHEET

```
dig options                    dig output
+dnssec   sets DO              flags: qr aa         header flags (sec. 1)
+norec    clears RD            EDNS: ... flags: do  OPT flags (sec. 2)
+cd       sets CD              udp: 1232            advertised UDP size
+adflag   sets AD in query     status:              RCODE (sec. 7)
+coflag   sets CO (9.20.11+)   TYPE128              NXNAME (older dig)
+tcp      use TCP

BIND queries.log flags, e.g. "+E(0)DK":
  +  RD set     E(0)  EDNS version 0     D  DO     C  CD
  K  cookie     T     TCP                S  signed (TSIG)
(Letters written from memory: check the BIND ARM.)
```